

Question 4

$$1. \quad \tilde{y}_i = \tilde{b}_0 + \tilde{b}_1 x_i$$

$$\tilde{y}_i = y_i - \bar{y}$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - \hat{y}_i)^2$$

$$\hookrightarrow \hat{b}_0 + \hat{b}_1 \tilde{x}_i$$

$$\frac{1}{n} \sum_{i=1}^n \tilde{y}_i = 0$$

$$\therefore 0 = \frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - (\hat{b}_0 + \hat{b}_1 \tilde{x}_i))^2$$

objective function: $\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - (\hat{b}_0 + \hat{b}_1 \tilde{x}_i))^2 + \alpha (\hat{b}_1)^2$

$$2. \quad \frac{d}{d\tilde{b}_0} \left[\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i)^2 + \alpha (\tilde{b}_1)^2 \right]$$

$$0 = -\frac{2}{n} \sum_{i=1}^n (\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i)$$

$$0 = \underbrace{\sum_{i=1}^n \tilde{y}_i}_{\substack{0 \\ \text{mean} \\ \text{normalization}}} - \underbrace{\sum_{i=1}^n \tilde{b}_0}_{\substack{0 \\ \text{mean} \\ \text{normalization}}} - \tilde{b}_1 \underbrace{\sum_{i=1}^n \tilde{x}_i}_{\substack{0 \\ \text{mean} \\ \text{normalization}}}$$

$$0 = \tilde{b}_0 n$$

$\tilde{b}_0 = 0$

$$\frac{d}{d\tilde{b}_1} \left[\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i)^2 + \alpha (\tilde{b}_1)^2 \right]$$

$$0 = \frac{1}{n} \sum_{i=1}^n 2(\tilde{y}_i - \cancel{\tilde{b}_0} - \tilde{b}_1 \tilde{x}_i)(-\tilde{x}_i) + 2\alpha \tilde{b}_1$$

$$= -\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i \tilde{x}_i - \tilde{b}_1 \tilde{x}_i^2) + 2\alpha \tilde{b}_1$$

$$= - \sum_{i=1}^n \tilde{y}_i \tilde{x}_i + \tilde{b}_1 \sum_{i=1}^n \tilde{x}_i^2 + n\alpha \tilde{b}_1$$

$$\tilde{b}_1 = \frac{\sum_{i=1}^n \tilde{y}_i \tilde{x}_i}{\sum_{i=1}^n \tilde{x}_i^2 + n\alpha}$$

3. The slope coefficient decreases as α increases

$$4. \frac{d}{d\tilde{b}_1} \left[\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i)^2 + \alpha |\tilde{b}_1| \right]$$

$$0 = \frac{1}{n} \sum_{i=1}^n 2(\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i)(-\tilde{x}_i) \pm \alpha$$

It becomes hard to take the derivative and optimize because we can end up with $+\alpha$ or $-\alpha$ depending on what $\tilde{b}_1$ is.

w/o penalty term

$$\frac{d}{d\tilde{b}_1} \left[\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i)^2 \right]$$

$$0 = \frac{1}{n} \sum_{i=1}^n 2(\tilde{y}_i - \tilde{b}_1 \tilde{x}_i)(-\tilde{x}_i)$$

$$\tilde{b}_1 = \frac{-\frac{2}{n} \sum_{i=1}^n \tilde{y}_i \tilde{x}_i}{\sum_{i=1}^n \tilde{x}_i^2}$$

↑
If this term's ($\tilde{b}_1$'s) magnitude is less than α , $\tilde{b}_1$ should be set to 0 as the term is no large enough to overcome the penalty imposed by α .